QUANSER
INNOVATE·EDUCATE

Mechatronic Sensors Trainer

Fundamental Labs – Load Cell

V1.0 – November 1st, 2025

Quanser Consulting Inc.
119 Spy Court
Markham, Ontario
L3R 5H6, Canada

info@quanser.com
Phone : 19059403575
Fax : 19059403576
printed in Markham, Ontario.

Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

Load Cell Lab

Why Explore Load Cells?

Load cells are fundamental tools in engineering for measuring force, weight, and pressure with high accuracy. Understanding how load cells work allows engineers to design systems that ensure safety, quality, and performance in applications such as structural testing, manufacturing, and automation. Their ability to provide precise, real-time measurements makes them essential for developing reliable and efficient engineering solutions.

Background

This lab is part of the Fundamentals labs for the Mechatronic Sensors Trainer. These labs will guide you through the sensors that are part of the board and the ones that are provided as external sensors so you can understand how to read from the sensors, what they output, when they are used, and be comfortable with using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**010_load_cell**

Getting Started

In this lab, you will use the **Sensors Trainer** to learn about the basics of **Load Cells**. The lab will have you explore the raw data from the sensor and get a feel of the Load Cell's capabilities and how to calibrate one. Finally, you will estimate the weight of an unknown object using the load cell.

Before you begin this lab, ensure that the following criteria are met:

- Make sure the load cell is set up as a scale. See the User Manual (Quanser/3_user_manuals\mech_sensors_trainer) section Assembling the Load Cell into a Scale (B.iii.) for instructions on this step.
- Make sure the Sensors Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/sensors_trainer) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
    - If using Python, we recommend using Visual Studio Code.